Helios Compute Control - illustrative venture investment committee memorandum

SYNTHETIC – NOT INVESTMENT ADVICE

Conditional posture: CONDITIONAL_INVEST subject to the executable terms and falsifiers below.

Authority: DECISION_RECORD_INCOMPLETE / PENDING_FOUNDER_SIGNATURE / no delegated investment authority.

Knowledge cutoff: 2026-08-29T00:00:00Z

Packet receipt: dc0bd679f21d67c728adf801eb0fc59ddcb15e317ddf9323be96b1983a958764

Recommendation and executable terms

Invest only if the 30% XIRR, 3.0x MOIC, and 10% modeled probability-below-1.0x hurdles clear and milestone funding stays tied to retention, pipeline, and margin evidence.

Binding loss hurdle: 20.00% (MC SE 1.26 pp) versus <=10%; status MISSES. The conditional posture is not funding approval while this test and the diligence gates remain open.

TERM	EXACT SELECTED-CASE POSITION
Security	Series C, 1x non-participating preference, senior to A/B and junior to any declared bridge
Initial / conditional capital	$25M at close / $15M at month 12
Valuation	$160M pre-money / $200M fully funded post-money
Series C ownership	13.51% at first close / 20.00% fully funded
Option pool	1,611,031 unissued shares after refresh; pre-money holders bear the refresh
Selected milestone returns	56.33% gross XIRR / 8.17x gross MOIC
Down-round returns	7.34% gross XIRR / 1.43x gross MOIC
Shortfall case	Bridge at month 36; 17.78% / 2.27x

Product, market, customers, competition, and business model

Helios can become the system of control for volatile enterprise GPU spend if ordinary cohorts retain and optimizer savings translate into durable platform economics.

DIMENSION	UNDERWRITING JUDGMENT	DISCONFIRMING FACT / OPEN GATE
Product	GPU-spend control plane with customer-level usage and cost records.	Production replication of the optimizer effect remains open.
Market	Tiered Bayesian survey supports a scenario range, not a market fact.	Tier 5 is data-thin and abstained; priors remain explicit.
Customers	Ordinary cohorts are separated from hand-picked design partners.	Pooled retention overstates repeatability.
Competition	Workflow value depends on neutral provider-cost control.	Provider-native tooling and cloud concentration may compress differentiation.
Business model	Usage-linked expansion can compound with customer compute spend.	Gross margin remains exposed to compute, telemetry, and support costs.
Terms	Milestone financing limits capital at risk before operating proof.	Failure creates explicit down-round or bridge dilution.

Counterthesis: Design-partner selection, inflated pipeline, cloud-cost exposure, and preference-heavy outcomes may make growth and TAM appear more durable than they are.

Team and financing accountability

These are role-level evidence states, not invented biographies or reference-checked conclusions.

DIMENSION	ASSESSMENT
Observable strengths	CEO / founder: Product insight is supported by design-partner problem definition. CTO: Optimizer experiment shows synthetic unit-cost improvement. CRO / revenue: Pipeline stage history exists at opportunity level. Finance: No dedicated capability is evidenced.
Unproven capabilities	CEO / founder: Commercial dependence and succession remain open. Evidence state: OPEN. CTO: Provider concentration and production replication remain open. Evidence state: PARTIAL. CRO / revenue: Ordinary-customer repeatability is not established. Evidence state: OPEN. Finance: Usage margin, runway, and financing controls need an accountable owner. Evidence state: ABSENT.
Key-person risk	OPEN - founder commercial dependence and succession evidence remain conditions to the second tranche.
Required capacity	Finance leader accountable for usage margin, monthly close, runway, and financing controls. Revenue operations owner accountable for ordinary-customer stage-to-close governance.

Cap table and financing-event bridge

SCENARIO	FUNDED SERIES C	OWNERSHIP	MINIMUM CASH	EXIT PROCEEDS	GROSS XIRR	GROSS MOIC
BASE	$25M	12.94%	$13.4M	$160.3M	44.97%	6.41x
MILESTONE	$40M	20.00%	$17M	$326.9M	56.33%	8.17x
DOWNSIDE	$25M	10.14%	$9M	$35.6M	7.34%	1.43x
FINANCING_SHORTFALL	$25M	10.79%	$250,000	$56.7M	17.78%	2.27x

SELECTED EVENT	MONTH	STATUS	CASH	NEW SHARES	FULLY DILUTED SHARES
series-c-close	1	FUNDED	$25M	1,814,223	13,425,254
series-c-tranche	12	FUNDED	$15M	1,088,534	14,513,788

Milestone financing and monthly runway

The $15M second tranche releases only when `ALL_TESTS_PASS_AFTER_CURE` after a 30-day cure. Evaluator: `BOARD_FINANCE_COMMITTEE`. Failure consequence: `WITHHOLD_TRANCHE_AND_REUNDERWRITE_RUNWAY`.

- `hx-nrr-metric` >= `1.05` over `trailing_12_months`.
- `hx-margin-metric` >= `0.70` over `trailing_3_months`.
- `hx-pipeline-audit` == `COMPLETE` over `as_of_test`.
- `hx-optimizer-replication` == `REPLICATED` over `pre_tranche`.

YEAR	ENDING CASH
1	$48.2M
2	$38.5M
3	$30.2M
4	$23M
5	$17M

Preference waterfall and investor return bridge

The selected exit equity value is not a naked outcome assumption. It is derived from the retained LTM revenue ledger and a declared five-year operating and valuation scenario:

EXIT-VALUE OPERAND	SELECTED MILESTONE CASE	CLASSIFICATION
Observed LTM revenue	$57M	Retained synthetic P&L
Annual revenue growth / hold	48.00% / 5 years	Scenario
Terminal revenue / exit multiple	$404.4M / 4.00x	Scenario
Exit enterprise value / cash	$1,617.6M / $17M	Accounting bridge; net debt is -$17M
Exit equity value	$1,634.6M	Waterfall operand

Exit value basis is `EQUITY_VALUE`. Unissued option-pool shares are treated as fully granted common at exit and share pro rata in residual proceeds. Class proceeds conserve to $1,634.6M with a 0-cent residual.

CLASS	ELECTION	PREFERENCE	RESIDUAL	TOTAL PROCEEDS
SERIES_A	CONVERT	$0	$230.9M	$230.9M
SERIES_B	CONVERT	$0	$208.4M	$208.4M
SERIES_C	CONVERT	$0	$326.9M	$326.9M

Series C invests $40M in dated funded tranches and receives $326.9M at exit: **56.33% gross XIRR / 8.17x gross MOIC**.

Conditional distribution and sensitivity

The 1,000 retained paths replay the full financing ledger and exact waterfall. They are scenario priors, not a forecast or evidence of real-world accuracy.

Declared scenario-state priors: Base 30.00%; Downside 15.00%; Financing Shortfall 10.00%; Milestone 45.00%.

STATISTIC	P10	P50	P90
Gross MOIC	0.00x	5.06x	9.92x
Gross XIRR	-100.00%	36.36%	61.82%

- Probability below 1.0x: **20.00%** (Monte Carlo SE **1.26 pp**, 1,000 draws).
- Sensitivity book: 13 full-engine cells across exit value, exit date, later-round price, and milestone state.

Econometric credit and zero-credit map

- HX-06 (`CAUSAL_SYNTHETIC_ONLY`) supports only the randomized synthetic optimizer estimand and tested population.
- HX-07 (`NOT_IDENTIFIED`) receives zero base-case causal credit because adoption follows spend spikes and identification fails.
- HX-05 (`PREDICTIVE_ASSOCIATION`) is a prior-sensitive market-size scenario, not causal evidence.

Value creation and board cadence

Combined full-model impact: **$3M minimum cash**, **$3.4M Series C proceeds**, **0.34% gross XIRR**, including an explicit **$0 interaction residual**.

INITIATIVE	EVIDENCE CLASS	BASELINE / TARGET	OWNER	MODELED CONSEQUENCE	STOP RULE / RISK
Ordinary-cohort expansion	HUMAN_JUDGMENT	117.07% / 125%	CRO	$75,504/month cash; $9.1M exit value; $1.5M cost	Design-partner tactics do not transfer
Optimizer unit economics	MIXED_CAUSAL_SYNTHETIC_AND_SCENARIO	70.61% / 74%	CTO	$53,661/month cash; $7.7M exit value; $2M cost	Provider price changes or workload mix invalidate transferred savings
Enterprise sales governance	DESCRIPTIVE	48 inflated opportunities / <15% forecast error	CRO / Finance	$0/month cash; $0 exit value; $1.2M cost	Enterprise cycle elongation remains unidentified

Economic mapping register

- **Ordinary-cohort expansion:** ordinary_base_arr_cents * max(0, target_nrr - observed_nrr) * gross_margin * 50% realization. Inputs: exit_multiple_on_annual_cash=10.0; gross_margin=0.7061223762551445; observed_nrr=1.1706504793654182; ordinary_base_arr_cents=3234114396; realization_factor=0.50; target_nrr=1.25
- **Optimizer unit economics:** ltm_compute_cost_cents * (1 - exp(precommitted_unadjusted_optimizer_itt_log_points)) * adoption_rate. Inputs: adoption_rate=0.65; causal_boundary=Synthetic ITT identifies the planted test population only; adoption and valuation multiple remain scenario judgments.; exit_multiple_on_annual_cash=12.0; ltm_compute_cost_cents=1138084029; multiplicative_cost_savings_rate=0.08704747623808384; precommitted_unadjusted_optimizer_itt_log_points=-0.09107139999999998
- **Enterprise sales governance:** zero_base_case_credit_until_stage_history_and_conversion_design_are_identified. Inputs: modeled_value_credit_cents=0

PHASE	TIMING	OWNER	MILESTONE	KPI	STOP RULE
Pre-close	Before signing	Deal lead	Verify cap table, preferences, side letters, and milestone definitions.	All financing operands reconciled	Do not fund with unresolved ownership or waterfall terms.
Day 1	Funding date	CEO / CFO	Lock usage-margin, runway, and ordinary-cohort definitions.	Signed operating metric dictionary	Do not release milestone capital on non-reconciled metrics.
Day 30	Close + 30 days	CFO	Install weekly runway and provider-cost forecast.	Cash, committed spend, gross margin	Escalate if runway falls below financing plan.
Day 100	Close + 100 days	CRO / CTO	Reconcile pipeline history and optimizer adoption economics.	Ordinary-cohort NRR, conversion, cost savings	Hold second tranche if milestones are not independently evidenced.
Year 1	Quarterly through month 12	Board	Re-underwrite ownership, runway, and exit cases.	Dilution, runway, margin, milestone state	Replan financing before the declared runway floor.

Falsifiers, open diligence, and limitations

- FALSIFIER: Ordinary-cohort NRR below 100%
- FALSIFIER: Pipeline conversion below 20%
- FALSIFIER: Gross margin below 65%
- FALSIFIER: Runway below 12 months post-close
- OPEN **HX-D01 · HIGH · PRE_IC** - Full opportunity stage-history export through the cutoff Owner: Revenue operations diligence lead. Decision consequence: Keep the conditional tranche withheld until conversion and weighted pipeline are recomputed from complete history.
- OPEN **HX-D02 · HIGH · PRE_SIGNING** - Design-partner contract and renewal sample Owner: Commercial diligence lead. Decision consequence: Re-underwrite ordinary-cohort transferability and remove pooled-retention credit if terms are not comparable.
- OPEN **HX-D03 · CRITICAL · PRE_TRANCHE** - Provider-level compute, telemetry, and support unit-cost ledger Owner: Technical and financial diligence leads. Decision consequence: Do not release milestone capital until gross-margin improvement reconciles to provider invoices and workload telemetry.
- OPEN **HX-D04 · CRITICAL · PRE_SIGNING** - Executed preference, pro-rata, option-pool, and side-letter schedule Owner: Deal counsel. Decision consequence: Do not fund until ownership and every waterfall scenario reconcile to executed terms.

This packet is generated from a fictional deterministic room. Synthetic causal estimates recover planted mechanisms only. Scenario outputs are not forecasts. The investment record remains unsigned and requires human IC authority.

Receipt appendix

- MILESTONE result: `39b567a968bc5754339d3416e68aa572e46c796fc9d4c27d7d4388cc7dd8a1e0`
- Sensitivity book: `a94fa30a09dd43d2cef42b61be8bb67052544972c97f8a0b0024458a6a8788d8`
- DOWNSIDE result: `bd68dbbc4a3fcc637b28930db53c8534c71d8e7a988ee294ae221be337e50484`
- Value-creation bridge: `56a8b0eb1f8872569600d18193809c78e7c4206fde3173f1587c40b5c066e4d9`